# Minjie Lei

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## Research Interests

Multiphase ISM structure; Galactic magnetism; CMB foreground; Astrostatistics

## Education

**Ph.D.**, Physics, Stanford University — Sept 2021 – Present
**Advisor**: Susan Clark
**Thesis**: Morphological Mapping of the Multiphase, Magnetized ISM

**B.Sc.**, Physics & Math, University of Michigan — Sept 2016 – June 2020
**Advisor**: James Wells
**Thesis**: "*Probing non-standard neutrino interactions with supernova neutrinos at Hyper-K.*"

## Publications

### First-Author

1. **Minjie Lei** et al., "*Super-resolving Polarized Dust Emission with Transformer-Based Multi-Tracer Fusion*, 2026, arXiv:2610.02575, Submitted to ApJ
2. **Minjie Lei** et al., "*Imprints of the Neutral Interstellar Medium on Polarized Synchrotron Emission and Faraday Rotation,* 2026, arXiv:2606.11359, Submitted to ApJ
3. **Minjie Lei**, S. E. Clark, et al., "*Neutral gas phase distribution from HI morphology: phase separation with scattering spectra and variational autoencoders,*" The Astrophysical Journal, vol. 993, no. 4, 2025
4. **Minjie Lei**, S. E. Clark, "*A New Constraint on the Relative Disorder of Magnetic Fields between Neutral ISM Phases,*" The Astrophysical Journal, vol. 972, no. 1, 2024
5. **Minjie Lei**, S. E. Clark, "*Probing the cold neutral medium through HI emission morphology with the scattering transform,*" The Astrophysical Journal, vol. 947, no. 2, 2023

### Other Publications

1. Iryna Butsky, Caleb Reshaw, **Minjie Lei** et al., "*Phase-dependent magnetic coherence in the turbulent interstellar medium,*" 2026, arXiv:2606.11323, Accepted by ApJ
2. Rebecca Booth, Anna Ordog et al. including **Minjie Lei**, "*A Three-Dimensional Model for the Reversal in the Local Large-Scale Interstellar Magnetic Field,*" The Astrophysical Journal, vol. 997, no. 2, 2026
3. Marc-Antoine Miville-Deschênes, JR Dawson et al. including **Minjie Lei**, "*The Multi-phase HI of the Milky Way and Nearby Galaxies,*" 2026, arXiv:2606.25089, Advancing Astrophysics with the SKA II (AASKAII)
4. **Minjie Lei**, K. V. Tsang, et al., "*Implicit Neural Representation as a Differentiable Surrogate for Photon Propagation in a Monolithic Neutrino Detector,*" 2022, arXiv:2211.01505

## Honors and Awards

2025–2026 **2MB Graduate Fellowship:** Awarded to a senior graduate student who has done developing statistical techniques for astrophysics, **Stanford University**

2023–2025 **Stanford Graduate Fellowship:** Awarded to graduate students in science and engineering who has demonstrated excellence in doctoral-level research and study, **Stanford University**

2017–2020 **James B. Angell Scholar**, **University of Michigan**

## Scientific Presentations

### Invited Talks

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|---|---|
| Sep 2026 | **CMB Probe for the 2030s Tagup**; NASA JPL |
| Oct 2025 | **Galaxy Seminar**; Flatiron CCA |
| Sep 2025 | **Low Density Universe Meeting**; STScI |
| Sep 2025 | **SFIR Seminar**; Princeton Astro |
| May 2025 | **Astronomy Tea Talk**; Caltech |
| April 2025 | **ITC Luncheon**; Havard CfA |
| March 2025 | **Scattering Club**; ENS Paris |
| Nov 2024 | **Astronominque Series**; Ciela Institute |
| March 2024 | **KIPAC tea**; Stanford University |
| Jan 2024 | **Journal Club**; Pan-Experiment Galactic Science Group |
| February 2023 | **ISM Seminar**; ENS Paris |

### Contributed Talks

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| Oct, 2026 | **Interstellar Institute 8**; nstitut Pascal - Université Paris-Saclay |
| May, 2026 | **AI+Science: Accelerating Discovery**; Stanford, CA |
| Jan, 2026 | **AAS 247**; Phoenix, Arizona |
| July, 2025 | **Interstellar Institute 7**; Institut Pascal - Université Paris-Saclay |
| Feb, 2025 | **Structure and polarization in the interstellar medium**; Stanford University |
| Jun. 2024 | **AAS 244**; Madison, Wisconsin |
| July. 2023 | **Interstellar Institute 6**; Institut Pascal - Université Paris-Saclay |
| Dec. 2022 | **Galactic Science and CMB Foregrounds**; Core to Core CMB Workshop series |

## Teaching and Mentorship

### Teaching Assistantships

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| Spring 2024 | **Computational Physics** (Physics 113) |
| Fall 2023 | **Stars and Planets in a Habitable Universe** (Physics 15) |
| Winter 2023 | **Stars and Planets in a Habitable Universe** (Physics 15) |
| Fall 2019 | **Honors Physics I - Mechanics** (Physics 160) |
| Fall 2018 | **Physics for the Life Sciences I** (Physics 135) |

### Student Mentorship

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| 2024-present | **Caio Gould**, undergraduate at Stanford |
| 2023-2024 | **Abraar Saleem**, Stanford postbac scholar, Stanford KIPAC postbac program |
| 2022-2023 | **Amritpal Nijjar**, undergraduate at UCLA, Stanford PIE Program |
| 2020-2021 | **Jackie Zhao**, High school student, Minds Matter Philadelphia |

## Service and Outreach

**Journal Reviewer:** A&A, MINRAS, Nature Astronomy
**Booth Leader & Volunteer**; KIPAC Community Day, Stanford University
**Local Organizing Committee**; Conference in Honor of Prof. John Dickey, Stanford University
**Booth Leader & Volunteer**; STEMFest, Stanford University
**Volunteer**; KIPAC Community Day, Stanford University

## References


**Susan Clark**, Stanford University seclark1@stanford.edu
**Erwan Allys**, ENS Paris erwan.allys@phys.ens.fr
**Doug Finkbeiner**, Harvard University dfinkbeiner@cfa.harvard.edu
**Marc-Antoine Miville-Deschênes**, CNRS & ENS Paris marc-antoine.miville-deschenes@cnrs.fr